Model Answer Assignment-3 EE322M

1. Consider the discrete-time sequence

$$x[n] = \cos\left(\frac{n\pi}{8}\right)$$

Find two different continuous-time signals that would produce this sequence when sampled at a frequency of $f_s = 10$ Hz.
Solution: Let us solve it for the more general case. Note that a signal $x_a(t) = \cos(2\pi f_0 t)$ that is sampled with a sampling frequency of f_s results in the discrete-time sequence

$$x[n] = x_a(nT_s) = \cos(2\pi f_0 n T_s) = \cos\left(2\pi \frac{f_0}{f_s} n\right), \text{ where } T_s = \frac{1}{f_s}.$$

For this question:

$$2\pi \frac{f_0}{f_s} n = \frac{\pi}{8} n \Rightarrow \frac{2f_0}{f_s} = \frac{1}{8} \Rightarrow f_0 = \frac{f_s}{16} = \frac{10}{16}.$$

Note that for any integer k,

$$\cos\left(2\pi \frac{f_0}{f_s} n\right) = \cos\left(2\pi \frac{f_0}{f_s} n + 2\pi k n\right) = \cos\left(2\pi \frac{f_0 + k f_s}{f_s} n\right). \tag{1}$$

The signal $\cos\left(2\pi \frac{f_0 + k f_s}{f_s} n\right)$ is the sampled version of the signal $\cos(2\pi(f_0 + k f_s)t)$ where the sampling frequency is f_s. From equation (1) it is clear that the samples of the signals $\cos(2\pi(f_0 + k f_s)t)$ and $\cos(2\pi f_0 t)$ are the same. Thus, two signals that have $\cos\left(\frac{n\pi}{8}\right)$ as the samples are:

$$x_1(t) = \cos\left(2\pi \frac{10}{16} t\right) \text{ and } x_2(t) = \cos\left(2\pi \left(\frac{10}{16} + 10\right) t\right).$$

2. The standard Fourier transform pair of a rectangular window of area AT centered at '0' time is given by,

$$A \operatorname{rect}\left(\frac{t}{T}\right) \leftrightarrow AT \operatorname{sinc}\left(\frac{\omega T}{2\pi}\right).$$

(See end of assignment on the exact definitions of rect and sinc). Using the above result, and the shifting, scaling and integration properties of Fourier transform, find the Fourier transform of the saw-tooth pulse shown in Figure 1.

Solution: The sawtooth pulse may be represented as a combination of two triangular pulses $x(t) = x_1(t) + x_2(t)$, where $x_1(t)$ represents the shifted triangular pulse of amplitude A. Hence, $x_2(t) = -x_1(-t)$ represents the shifted triangular pulse of amplitude $-A$.

If $x_3(t) = A\ \texttt{tri}\left(\frac{t}{2T}\right)$ represents a triangular pulse of area AT, and centered at '0' time, then, $x_1(t) = x_3(t + T)$.

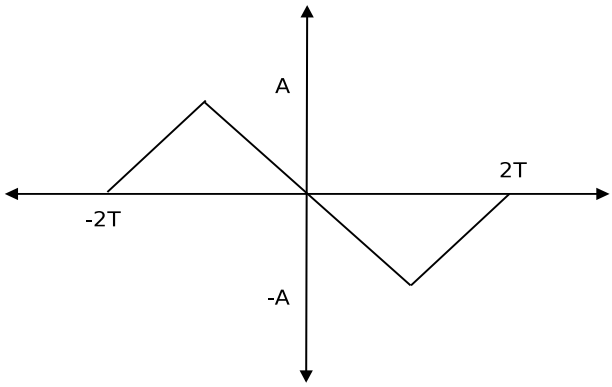


Figure 1: Sawtooth pulse.

Let us consider a rectangular window $y(t)$ of area AT, and centered at '0' time.

$$y(t) = A\ \texttt{rect}\left(\frac{t}{T}\right) \leftrightarrow Y(\omega) = AT\ \texttt{sinc}\left(\frac{\omega T}{2\pi}\right)$$

The signal $y_1(t)$ may be represented as,

$$y_1(t) = \frac{1}{T}y\left(t + T/2\right) - \frac{1}{T}y\left(t - T/2\right)$$

The signals $x_3(t)$, $y(t)$ and $y_1(t)$ are shown in figure 2. Hence,

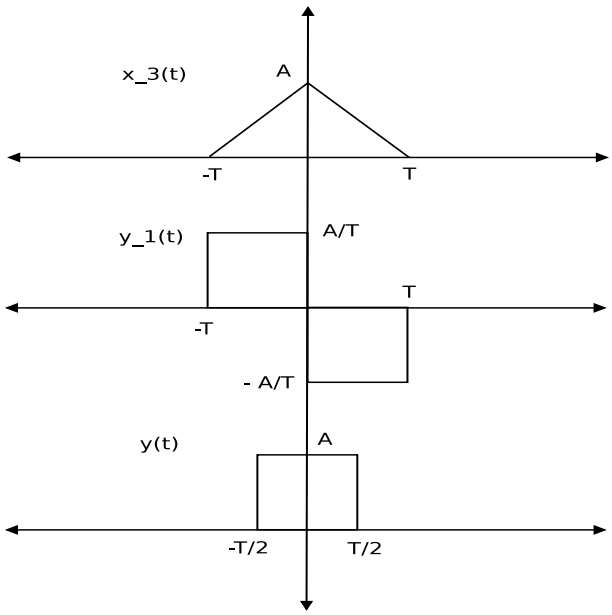


Figure 2: $x_3(t)$, $y_1(t)$ **and** $y(t)$.

$$\begin{aligned} Y_1(\omega) &= \frac{1}{T}e^{j\omega T/2}Y(\omega) - \frac{1}{T}e^{-j\omega T/2}Y(\omega) = \frac{1}{T}Y(\omega)\left[e^{j\omega T/2} - e^{-j\omega T/2}\right] \\ &= \frac{2j}{T}Y(\omega)\sin(\omega T/2) = 2Aj\sin\left(\frac{\omega T}{2}\right)\text{sinc}\left(\frac{\omega T}{2\pi}\right) \end{aligned}$$

Again, $x_3(t) = \int_{-\infty}^{t} y_1(\tau)d\tau$. Hence, by the integration property of FT, we have

$$\begin{aligned} X_3(\omega) &= \frac{1}{j\omega}Y_1(\omega) + \pi Y_1(j0)\delta(\omega) = \frac{2A}{\omega}\sin\left(\frac{\omega T}{2}\right)\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \\ &= \frac{AT}{wT/2}\sin\left(\frac{\omega T}{2}\right)\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) = AT\frac{\sin\left(\pi\frac{\omega T}{2\pi}\right)}{\pi\frac{\omega T}{2\pi}}\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right) \\ &= AT\left[\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)\right]^2. \end{aligned}$$

Hence,

$$\begin{aligned} x_1(t) = x_3(t+T) \leftrightarrow X_1(\omega) &= e^{j\omega T}X_3(\omega) = e^{j\omega T}AT\left[\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)\right]^2 \\ x_2(t) = -x_1(-t) \leftrightarrow X_2(\omega) &= -X_1(-\omega) = -e^{-j\omega T}AT\left[\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)\right]^2. \end{aligned}$$

Hence,

$$\begin{aligned} x(t) = x_1(t) + x_2(t) \leftrightarrow X(j\omega) &= [e^{j\omega T} - e^{-j\omega T}]AT\left[\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)\right]^2 \\ &= 2j\sin(\omega T)\,AT\left[\operatorname{sinc}\left(\frac{\omega T}{2\pi}\right)\right]^2. \end{aligned}$$

The magnitude and phase plots of the sawtooth pulse, with A=1 and T=1, is shown in figure 3.

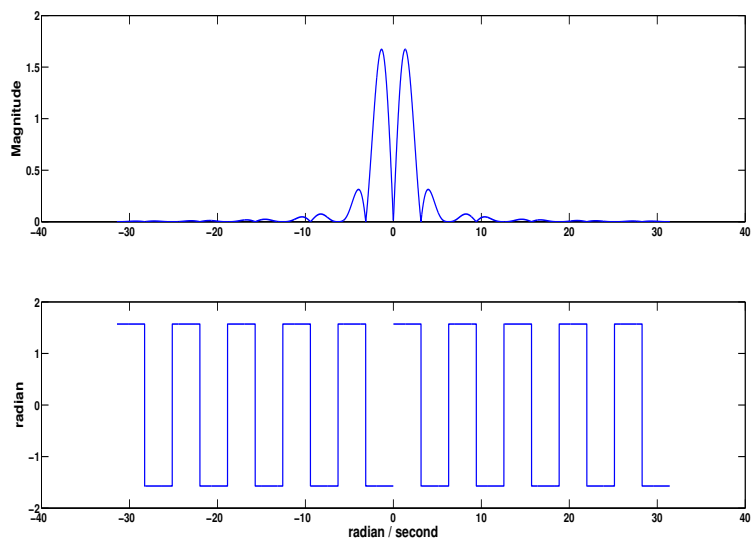


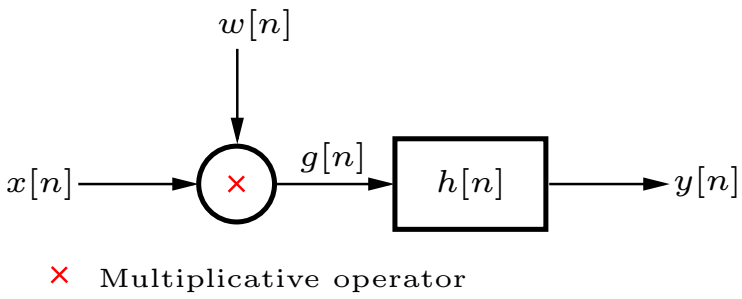
Figure 3: **Magnitude and Phase Response of a sawtooth pulse.**

3. Consider the discrete -time system given in the block diagram below:

   Let $h[n] = \dfrac{sin(\frac{\pi n}{2})}{\pi n}$ be the impulse response of the system and the input $x[n]$ is corrupted by a multiplicative noise of the form $w[n] = (-1)^n$.

   (a) Sketch $H(e^{j\omega})$ and $G(e^{j\omega})$.

   (b) Determine the output $y[n]$ for input $x[n] = \dfrac{sin(\frac{\pi n}{4})}{\pi n}$.



Solution: Given,

$$y[n] = (x[n] \times w[n]) * h(n) = g[n] * h[n]$$

Given

$$h[n] = \frac{sin(\frac{\pi n}{2})}{\pi n}$$

We have DTFT pairs,

$$x[n] = \frac{sin(\frac{\pi n}{4})}{\pi n} \quad \xrightarrow{DTFT} \quad X(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |\omega| < \pi \end{cases}$$

$$w[n] = (-1)^n = e^{-j\pi n} \quad \xrightarrow{DTFT} \quad W(e^{j\omega}) = 2\pi\delta(\omega - \pi)$$

We have,

$$g[n] = x[n] \times w[n] \quad \xrightarrow{DTFT} \quad G(e^{j\omega}) = \frac{1}{2\pi}\left(X(e^{j\omega}) * W(e^{j\omega})\right).$$

Therefore,

$$G(e^{j\omega}) = \frac{1}{2\pi} \times 2\pi \times \delta(\omega - \pi) * X(e^{j\omega}) = X\left(e^{j(\omega-\pi)}\right) = \begin{cases} 1, & |\omega - \pi| \leq \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |\omega - \pi| < \pi \end{cases}$$

$$= \begin{cases} 1, & \frac{3\pi}{4} \leq \omega \leq \frac{5\pi}{4} \\ 0, & 0 < \omega < \frac{3\pi}{4} \text{ and } \frac{5\pi}{4} < \omega < 2\pi. \end{cases}$$

$G(e^{j\omega})$ is periodic with period 2π. $G(e^{j\omega})$ is plotted as shown in Figure 4(a).
Further,

$$h[n] = \frac{sin(\frac{\pi n}{2})}{\pi n} \xrightarrow{DTFT} H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} \leq |\omega| < \pi \end{cases}.$$

$H(e^{j\omega})$ is periodic with 2π. $H(e^{j\omega})$ is plotted as shown in Figure 4(b).
Since

$$y[n] = g[n] * h[n] \quad \xrightarrow{DTFT} \quad Y(e^{j\omega}) = G(e^{j\omega})H(e^{j\omega})$$

and $G(e^{j\omega})$ and $H(e^{j\omega})$ are non overlapping rectangular functions, $Y(e^{j\omega}) = 0 \Rightarrow y[n] = 0$.

4. Find the signal energy of

$$x[n] = \text{sinc}\left(\frac{n}{10}\right) \sin\left(\frac{2\pi n}{4}\right).$$

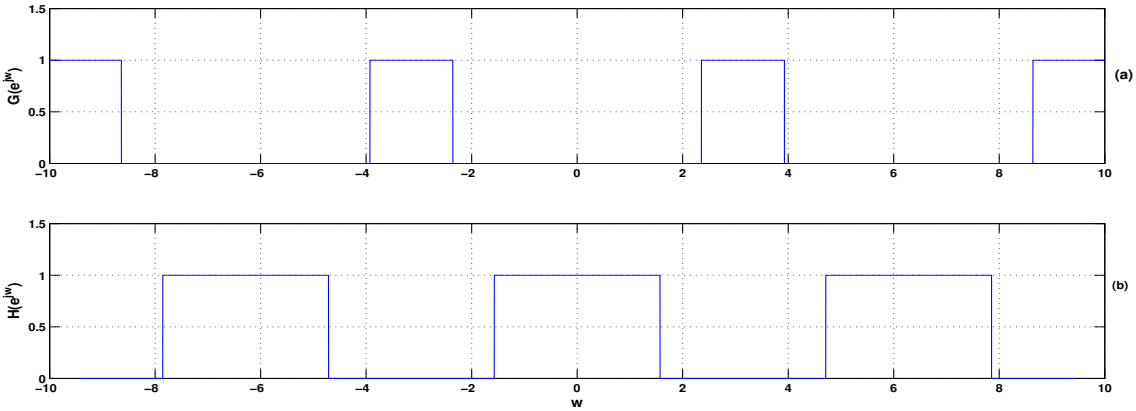


Figure 4: (a) $G(e^{j\omega})$ and (b) $H(e^{j\omega})$

(Use the sinc definition that you have used for Question 2).
Solution: The energy of a signal is defined as

$$E_x = \sum_{-\infty}^{\infty} |x(n)|^2$$

We can avoid doing a complicated infinite summation by using Parseval's theorem. It is given by

$$\sum_{-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi} |X(e^{j\omega})|^2 d\omega = \int_0^1 |X(f)|^2 df.$$

The DTFT of first term in $x[n]$ can be found by starting with the Fourier pair

$$\text{sinc}\left(\frac{n}{10}\right) \xrightarrow{DTFT} 10\,\text{rect}(10f) * \delta(f).$$

DTFT of second term in $x[n]$ can be found by the Fourier pair

$$\sin\left(\frac{2\pi n}{4}\right) \xrightarrow{DTFT} \frac{j}{2}\left[\delta\left(f+\frac{1}{4}\right) - \delta\left(f-\frac{1}{4}\right)\right].$$

Using the multiplication-convolution duality of the DTFT

$$\begin{aligned}
\text{sinc}\left(\frac{n}{10}\right) \times \sin\left(\frac{2\pi n}{4}\right) &\xrightarrow{DTFT} \{10\,\text{rect}\,(10f) * \delta(f)\} * \left\{\frac{j}{2}\left[\delta\left(f+\frac{1}{4}\right) - \delta\left(f-\frac{1}{4}\right)\right]\right\} \\
&= 5j\,\text{rect}(10f) * \left[\delta\left(f+\frac{1}{4}\right) - \delta\left(f-\frac{1}{4}\right)\right] \\
&= 5j\left[\text{rect}\left(10\left(f+\frac{1}{4}\right)\right) - \text{rect}\left(10\left(f-\frac{1}{4}\right)\right)\right].
\end{aligned}$$

Applying Parseval's theorem, we get

$$\begin{aligned}
E_x &= \int_0^1 \left| j5\left[\text{rect}\left(10\left(f+\frac{1}{4}\right)\right) - \text{rect}\left(10\left(f-\frac{1}{4}\right)\right)\right]\right|^2 df \\
&= 25\int_0^1 \left|\left[\text{rect}\left(10\left(f+\frac{1}{4}\right)\right) - \text{rect}\left(10\left(f-\frac{1}{4}\right)\right)\right]\right|^2 df.
\end{aligned}$$

The square of the sum equals the sum of the squares because there is no cross product as the two rectangles do not overlap.

$$\begin{aligned}
E_x &= 25\left\{ \int_0^1 \left[\text{rect}\left(10\left(f+\frac{1}{4}\right)\right)\right]^2 df + \int_0^1 \left[\text{rect}\left(10\left(f-\frac{1}{4}\right)\right)\right]^2 df \right\} \\
&= 25\left\{ \int_{\frac{-1}{4}-\frac{1}{20}}^{\frac{-1}{4}+\frac{1}{20}} 1\, df + \int_{\frac{1}{4}-\frac{1}{20}}^{\frac{1}{4}+\frac{1}{20}} 1\, df \right\} \\
&= 25\left(\frac{1}{10}+\frac{1}{10}\right) \\
&= 5.
\end{aligned}$$

5. Let x(t) be a real signal with the Fourier transform $X(j\omega)$. Suppose we are given the following,

   (a) $x(t)$ is real.

   (b) $x(t) = 0$ for $t \leq 0$.

   (c) $\frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}\{X(j\omega)\}e^{j\omega t}dw = |t|e^{-|t|}$.

   Determine the closed form expression for $x(t)$.
   Solution: Let $x(t) = x_e(t) + x_o(t)$, where $x_e(t)$ and $x_o(t)$ are the even and odd parts of $x(t)$. Since $x(t)$ is real, we have

$$x_e(t) \xrightarrow{CTFT} \texttt{Re}\{X(j\omega)\} \quad \text{and} \quad x_o(t) \xrightarrow{CTFT} j\texttt{Im}\{X(j\omega)\}.$$

   Given

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} Re\{X(j\omega)\}e^{j\omega t}d\omega = |t|e^{-|t|} \Rightarrow \texttt{IFT}\{Re\{X(j\omega)\}\} = |t|e^{-|t|} \Rightarrow x_e(t) = |t|e^{-|t|}.$$

   Thus,

$$x_e(t) = |t|e^{-|t|} \Rightarrow \frac{x(t)+x(-t)}{2} = |t|e^{-|t|}. \tag{2}$$

   Since $x(t) = 0$ for $t \leqslant 0$, we have $x(-t) = 0$ for $t \geqslant 0$. Using this in equation (2), we have

$$x(t) = 2|t|e^{-|t|} \quad \text{for} \quad t \geq 0 \Rightarrow x(t) = 2te^{-t}u(t).$$

6. A three-point symmetric moving average, referred to as a weighted moving average, is of the form

$$y[n] = b(ax[n-1] + x[n] + ax[n+1]).$$

   (a) Determine, as a function of a and b, the frequency response $H(e^{j\omega})$ of the above mentioned three-point moving average.

   (b) Determine the scaling factor b such that $H(e^{j\omega})$ has unity gain at zero frequency.

   (c) In many time-series analysis problems, a common choice for the coefficient a in the weighted moving average is $a = 0.5$. Determine and sketch the frequency response of the resulting filter.

   Solution:

(a) The frequency response for the above system is given as:

$$Y(e^{j\omega}) = b(ae^{-j\omega} + 1 + ae^{j\omega})X(e^{j\omega})$$
$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = b(ae^{-j\omega} + 1 + ae^{j\omega})$$
$$H(e^{j\omega}) = b(1 + a[e^{j\omega} + e^{-j\omega}]) = b(1 + 2a\cos\omega)$$

(b) At zero frequency, $H(e^{j0}) = b(1 + 2a)$. Therefore, $b(1 + 2a) = 1 \Rightarrow b = \frac{1}{(1+2a)}$.

(c) If $a = 0.5$, then $b = 0.5$. Therefore $H(e^{j\omega}) = \frac{1}{2}[1 + \cos\omega]$. The frequency response is plotted in Figure 5.

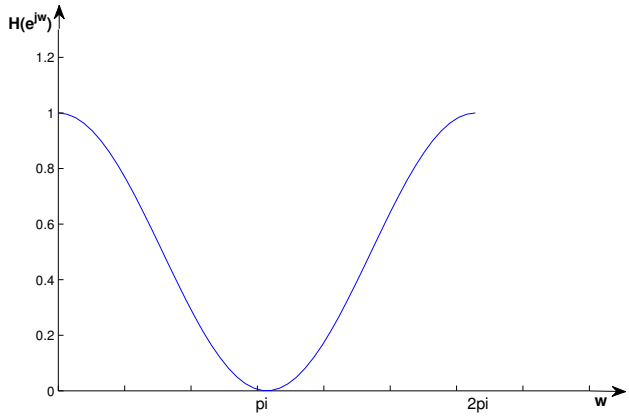


Figure 5: Magnitude and Phase plot of $H(e^{j\omega})$

7. A continuous-time signal $x_a(t)$ is to be filtered to remove frequency components in the range 5 kHz $< |f| <$ 10 kHz. The maximum frequency present in $x_a(t)$ is 20 kHz. The filtering is to be done by sampling $x_a(t)$, filtering the sampled signal, and reconstructing an analog signal using an ideal D/C converter.

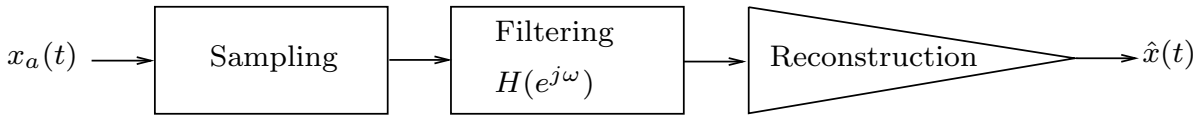


Figure 6: Figure for question 7

Find the minimum sampling frequency that may be used to avoid aliasing, and for this minimum sampling rate, find the frequency response of the ideal digital filter $H(e^{j\omega})$ that will remove the desired frequencies from $x_a(t)$.

Solution: The highest frequency of $x_a(t)$ is 20 kHz. Hence, the minimum sampling frequency to avoid aliasing is $f_s = 40$ kHz.

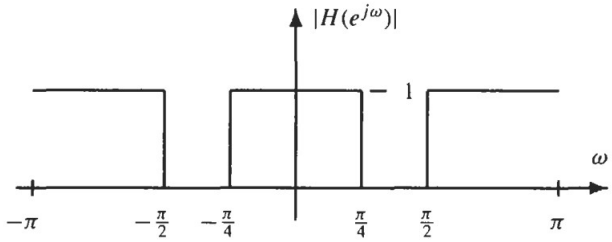
From solution of question (1) it is clear that the relation between continuous frequency of a signal and the digital frequency of the sampled version of the signal is

$$\omega = 2\pi\frac{f}{f_s}, \text{ where } \omega \text{ is the frequency of the sampled signal.}$$

Therefore, the frequency range 5 kHz $\leqslant f \leq$ 10 kHz corresponds to a digital frequency range

$$\frac{\pi}{4} \leqslant \omega \leqslant \frac{\pi}{2}.$$

and the desired filter is a band-stop filter that has a frequency response as illustrated in the figure below.



8. A continuous-time filter has a system function

$$H_a(s) = \frac{1}{s+1}.$$

If the impulse response of the above filter $h_a(t)$ is sampled to form a discrete-time system with a unit sample response $h[n] = h_a(nT_s)$, then find the value for T_s so that $H(e^{j\omega})$ at $\omega = \pi/2$ is down 6 dB from its maximum value at $\omega = 0$.

Solution: The impulse response of the continuous system is $h_a(t) = e^{-t}u(t)$. When sampled with a sampling period T_s, the resulting unit sample response is $h[n] = h_a(nT_s) = e^{-nT_s}u[n]$ and the frequency response is

$$H\left(e^{j\omega}\right) = \sum_{n=0}^{\infty} e^{-nT_s}e^{-jn\omega} = \sum_{n=0}^{\infty} e^{-(T_s+j\omega)n} = \frac{1}{1-e^{-T_s}e^{-j\omega}}.$$

With

$$|H(e^{j0})| = \frac{1}{1-e^{-T_s}} \quad \text{and} \quad |H(e^{j\pi/2})| = \frac{1}{|1+e^{-T_s}e^{-j\pi/2}|} = \frac{1}{|1-je^{-T_s}|} = \frac{1}{\sqrt{1+e^{-2T_s}}}$$

We want the following:

$$\begin{aligned}
&20\log|H(e^{j0})| - 20\log|H(e^{j\pi/2})| = 6 \\
\Rightarrow\ &20\left(\log\left(\frac{1}{1-e^{-T_s}}\right) - \log\left(\frac{1}{\sqrt{1+e^{-2T_s}}}\right)\right) = 6 \\
\Rightarrow\ &10\log\left(\frac{1+e^{-2T_s}}{(1-e^{-T_s})^2}\right) = 6.
\end{aligned}$$

Let $e^{-T_s} = x$. Then we have

$$\frac{1+x^2}{(1-x)^2} = 10^{0.6} \Rightarrow 1+x^2 = 3.981(1+x^2-2x) \Rightarrow 2.981x^2 - 7.962x + 2.981 = 0$$

Therefore,

$$\begin{aligned}
&x = 2.22058 \Rightarrow e^{-T_s} = 2.22058 \Rightarrow T_s = -0.7978 \ (T_s \text{ cannot be negative}) \\
&x = 0.450332 \Rightarrow e^{-T_s} = 0.450332 \Rightarrow T_s = 0.7978\,\texttt{sec}.
\end{aligned}$$

9. Determine whether each of the following statements is true or false. Justify your answers.

(a) An odd and imaginary signal always has an odd and imaginary Fourier transform.
Solution: First we check the Fourier transform of a real and odd function $x(t)$.

$$\begin{aligned}
X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt &= \int_{-\infty}^{\infty} x(t)\left(\cos(\omega t) - j\sin(\omega t)\right)dt \\
&= \int_{-\infty}^{\infty} x(t)\cos(\omega t)dt - j\int_{-\infty}^{\infty} x(t)\sin(\omega t)dt \qquad (3)
\end{aligned}$$

We evaluate the integral $\int_{-\infty}^{\infty} x(t)\cos(\omega t)dt$. Note that

$$\int_{-\infty}^{\infty} x(t)\cos(\omega t)dt = \int_{-\infty}^{0} x(t)\cos(\omega t)dt + \int_{0}^{\infty} x(t)\cos(\omega t)dt$$

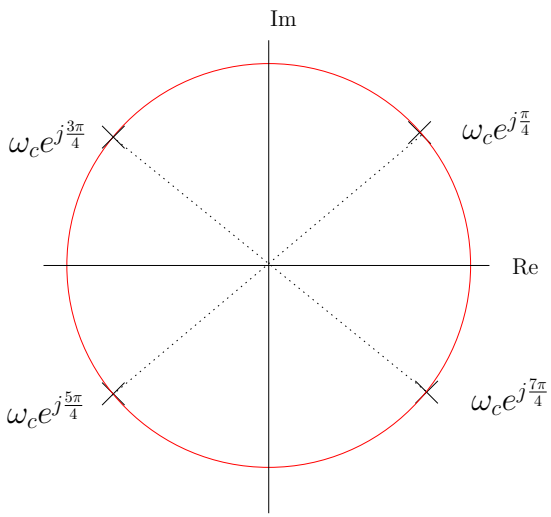
Substituting $-t$ for t, we have

$$\begin{aligned}\int_{-\infty}^{\infty} x(t)\cos(\omega t)dt &= \int_{-\infty}^{0} -x(-t)\cos(-\omega t)dt - \int_{0}^{\infty} x(-t)\cos(-\omega t)dt \\ &= \int_{-\infty}^{0} -x(t)\cos(\omega t)dt - \int_{0}^{\infty} x(t)\cos(\omega t)dt \\ &= \int_{0}^{\infty} x(t)\cos(\omega t)dt - \int_{0}^{\infty} x(t)\cos(\omega t)dt = 0.\end{aligned}$$

Using this in equation (3), we have $X(j\omega) = -j\int_{-\infty}^{\infty} x(t)\sin(\omega t)dt$.
Now let $y(t) = jx(t)$; this is an odd and purely imaginary signal. The Fourier transform of this signal is $Y(j\omega) = jX(j\omega) = \int_{-\infty}^{\infty} x(t)\sin(\omega t)dt$. Thus the Fourier transform of $y(t)$ is real. Hence the statement is false.

(b) The convolution of an odd Fourier transform with an even Fourier transform is always odd.
Solution: The Fourier transform of an odd signal is odd and that of a real signal is real. Since convolution in frequency domain is equivalent to multiplication in time domain, convolution of an odd Fourier transform with an even Fourier transform is equivalent to multiplication of an odd signal with a even signal. The result is an odd signal. Hence the statement is true.

10. A fourth-order Butterworth filter with cut-off frequency ω_c has the following pole location:



This implies that a Butterworth filter is not causal. If it is not causal, then we cannot implement such a filter using a circuit. Find out the error in the reasoning and justify why Butterworth filters are causal and can be implemented using a circuit.
Solution: All the points made in the question are wrong.

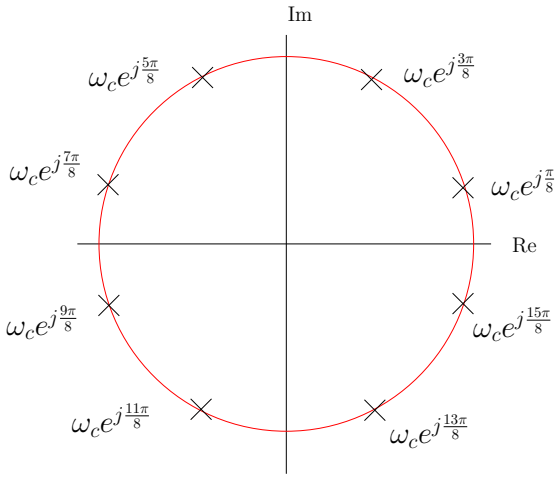
(a) *Pole locations of the filter:* Butterworth filters do not have poles like the one shown in the figure. If $B(j\omega)$ is the frequency response of an N-th order Butterworth filter, then

$$|B(j\omega)|^2 = \frac{B_0^2}{1 + (\frac{j\omega}{j\omega_c})^{2N}}, \text{ where } B_0 \text{ is the DC gain.}$$

Thus for a fourth-order Butterworth filter:

$$|B(j\omega)|^2 = \frac{B_0^2}{1+(\frac{j\omega}{j\omega_c})^8}, \text{ where } B_0 \text{ is the DC gain.}$$

Hence the pole locations of $|B(j\omega)|^2$ are as shown in the figure below



Note that these are not the pole locations of the filter ($B(j\omega)$) but that of the squared magnitude of the frequency response ($|B(j\omega)|^2$).
Further, $|B(j\omega)|^2 = B(j\omega) \times B^*(j\omega) = B(j\omega) \times B(-j\omega)$. Now we try to find the transfer function $B(s)$ of the Butterworth filter. If we consider $s = j\omega$, we have

$$|B(s)|^2 = B(s)B(-s) = \frac{B_0^2}{1+(\frac{s^8}{\omega_c^8})} = \frac{B_0^2\omega_c^8}{s^8+\omega_c^8}.$$

The roots of $s^8+\omega^8$ are the ones shown in the figure above. Our aim is to find $B(s)$, the transfer function of the Butterworth filter. Since $s^8+\omega^8$ have eight roots it can be factored in the following ways:

$$\begin{aligned} s^8+\omega^8 =& (s+\omega_c e^{j\frac{\pi}{8}})(s+\omega_c e^{j\frac{3\pi}{8}})(s+\omega_c e^{j\frac{5\pi}{8}})(s+\omega_c e^{j\frac{7\pi}{8}}) \\ & (s+\omega_c e^{j\frac{9\pi}{8}})(s+\omega_c e^{j\frac{11\pi}{8}})(s+\omega_c e^{j\frac{13\pi}{8}})(s+\omega_c e^{j\frac{15\pi}{8}}) \end{aligned}$$

Rearranging the factors we can write

$$\begin{aligned} s^8+\omega^8 = & \underbrace{(s+\omega_c e^{j\frac{5\pi}{8}})(s+\omega_c e^{j\frac{7\pi}{8}})(s+\omega_c e^{j\frac{9\pi}{8}})(s+\omega_c e^{j\frac{11\pi}{8}})}_{\texttt{denominator of } B(s)} \\ & \underbrace{(s+\omega_c e^{j\frac{\pi}{8}})(s+\omega_c e^{j\frac{3\pi}{8}})(s+\omega_c e^{j\frac{13\pi}{8}})(s+\omega_c e^{j\frac{15\pi}{8}})}_{\texttt{denominator of } B(-s)}. \end{aligned}$$

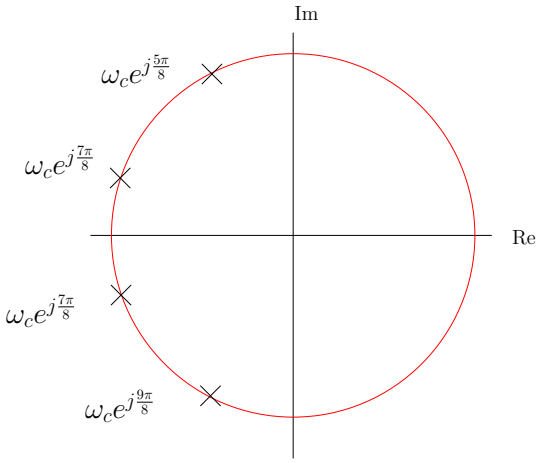
Thus the frequency response of the Butterworth filter is

$$B(s) = \frac{B_0\omega_c^4}{(s+\omega_c e^{j\frac{5\pi}{8}})(s+\omega_c e^{j\frac{7\pi}{8}})(s+\omega_c e^{j\frac{9\pi}{8}})(s+\omega_c e^{j\frac{11\pi}{8}})}$$

Note that

$$\begin{aligned} B(-s) &= \frac{B_0\omega_c^4}{(-s+\omega_c e^{j\frac{5\pi}{8}})(-s+\omega_c e^{j\frac{7\pi}{8}})(-s+\omega_c e^{j\frac{9\pi}{8}})(-s+\omega_c e^{j\frac{11\pi}{8}})} \\ &= \frac{B_0\omega_c^4}{(s-\omega_c e^{j\frac{5\pi}{8}})(s-\omega_c e^{j\frac{7\pi}{8}})(s-\omega_c e^{j\frac{9\pi}{8}})(s-\omega_c e^{j\frac{11\pi}{8}})} \\ &= \frac{B_0\omega_c^4}{(s+\omega_c e^{j\frac{13\pi}{8}})(s+\omega_c e^{j\frac{15\pi}{8}})(s+\omega_c e^{j\frac{\pi}{8}})(s+\omega_c e^{j\frac{3\pi}{8}})} \end{aligned}$$

Thus the poles of the Butterworth filter with transfer function $B(s)$ are on the left-half of the $\mathbb{C}$ plane as shown in the figure below.



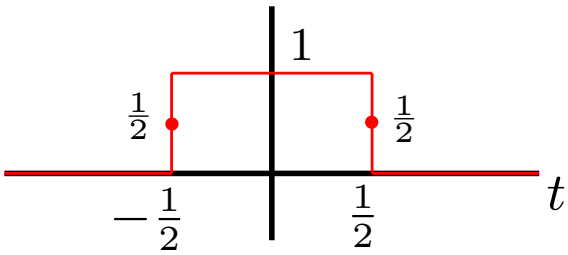
In general, the roots of an N-th order Butterworth filter are located at $s_k = \omega_c e^{\frac{j\pi}{2N}(2k+N-1)}$, where $k = 0, 1, 2, \ldots, N$.

(b) *Causality and pole location:* A causal system can have poles on the right-half $\mathbb{C}$ plane. For example, a system with impulse response $h(t) = e^t u(t)$ have a transfer function $H(s) = \frac{1}{s-1}$ and hence poles on the right-half of the s-plane. Hence pole locations do not divulge information about causality. As a matter of fact a Butterworth as shown above is stable. We need to further impose the condition of causality for the filter to be causal. Filters that deal with time-domain signals in real time have to be causal.

`Information:`

- `The rectangular function, also known as the gate function, is defined as follows:`

$$\text{rect}\,(t) = \Pi(t) = \begin{cases} 0, & |t| > \frac{1}{2} \\ \frac{1}{2}, & |t| = \frac{1}{2} \\ 1, & |t| < \frac{1}{2} \end{cases}$$



- `The` sinc `function is defined in two ways: The historical way of defining` sinc `function is:`

$$\text{sinc}\,(x) = \begin{cases} 1, & x = 0, \\ \dfrac{\sin(x)}{x}, & x \neq 0. \end{cases}$$

`This is the way I have defined` sinc `in one of our modules (Module-8b page-6). It is important to know the other way of defining it too -- since you might encounter it in some text or paper while reading.`

$$\text{sinc}\,(x) = \begin{cases} 1, & x = 0, \\ \dfrac{\sin(\pi x)}{\pi x}, & x \neq 0. \end{cases}$$

Both these definitions of sinc are extensively used in the literature. But note that these are just symbols and you need to decipher which definition is being used in Problem 2. You already know how rectangular window and the sinc function are related by Fourier transform (Module-8b). Just check which definition fits in to the result that you already know (Example A, Page-3, Module-8b).

- Notice in Problem 8, the system's impulse response is being sampled. We have studied sampling from the point of view of signals, i.e. signals being sampled to get discrete-time signal. However, if we need to find the discretized version of a system, we can sample the impulse response of the the continuous-time system to get the discrete-time version of the system. This is the idea behind the construction of digital Butterworth and Chebyshev filters illuded to in page 12 Module-10c.